

Julien TANG

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in [julien-tang](https://www.julien-tang.com)

I am a PhD student in cosmology at Centre Pierre Binétruy, a joint International Research Laboratory between UC Berkeley and CNRS.

I am interested in data simulation and data analysis pipelines for current and next-generation CMB experiments, such as Simons Observatory. This includes mapmaking, foreground separation and removal of the lensing signal in the quest for primordial gravitational waves.

Education

Université Paris Cité & Lawrence Berkeley National Laboratory

Oct 2024 – present

PhD in Cosmology

Pipeline development for next-generation CMB surveys, with Jacques Delabrouille and Michel Piat at AstroParticle and Cosmology laboratory and Centre Pierre Binétruy

ETH Zürich

Sept 2023 – Sept 2024

High-Energy Physics Master

Master thesis: *Mapmaking methods comparison for CMB observations from Chile*, with Alexandre Refregier and Jacques Delabrouille at Lawrence Berkeley National Laboratory

École polytechnique

Sept 2020 – Sept 2023

Cycle Ingénieur polytechnicien

Senior thesis: *Reconstruction of Inflationary CMB B-mode Signal in the Presence of Instrumental Effects and Galactic Foreground Emissions*, with Tomotake Matsumura and Josquin Errard at Kavli IPMU (U. Tokyo)

Lycée Sainte-Geneviève

Sept 2018 – Sept 2020

Preparatory courses for Grandes Écoles (MPSI/MP)*

Competitive undergraduate program in Maths and Physics, training students for enrolment in one of the Grandes Écoles (French Graduate School of Science in Engineering)

École Massillon

Sept 2015 – June 2018

High school

French baccalauréat : With highest honors.

Fellowships & Awards

Student Fellowship

Sept 2024 – Sept 2027

Lawrence Berkeley National Laboratory

Full PhD scholarship

Sept 2024 – Sept 2027

École Doctorale de l'Institut Polytechnique de Paris

Erasmus+ Mobility grant

Feb 2023 – Aug 2023

École Polytechnique, program KA171

Winning team of the International Physicists' Tournament's 14th edition

May 2022

1st out of 15 countries, Universidad Industrial de Santander, Bucaramanga Colombia,

[IPT 14th edition Colombia](#)

Collaboration membership

CMB-S4

since March 2024

Member of the Berkeley Lab CMB group, working on Data Management

Simons Observatory

since October 2025

Junior member, working on delensing, data synthesis and data reduction

Teaching & Outreach

Teaching

- Oral examiner (khôlleur) in Physics –Lycée Blaise Pascal, Orsay, Sept 2022 – Feb 2023
- English tutor – École Polytechnique, Sep 2021 – June 2022

Outreach

- UC Berkeley Astro Night speaker, Spring 2025
 - Opération Monge –École Polytechnique , 2020 – 2021
- Outreach in French high schools to encourage high school students, especially young women and minorities, to pursue studies in STEM

Contributed talks

- *Mapmaking for next-generation CMB surveys* (Fireslide), CMB-S4 Collaboration Meeting, March 2025
- *Mapmaking pipelines comparison from Chile for CMB-S4* – CMB-France #6, December 2024
- *Parametric component separation in the presence of beam systematics* – ILANCE Student-Fest #6, June 2023

Skills

Programming Languages

Python (full proficiency), Julia, Java.

Packages for CMB data analysis

Standard libraries (numpy, scipy, healpy, sympy, jax, mpi4py)
TOAST, a software framework for simulation and analysis of CMB-observations.

Delensalot, Curved-sky optimal CMB lensing reconstruction and bias calculation.

FGBuster, Parametric component separation for CMB observations.

Spoken languages

Bilingual in French and English (TOEFL iBT: 115/120)

German C1 level (DeutschSprachDiplom II obtained in 2018)

Intermediate level in Mandarin, Notions of Japanese

Tools

L^AT_EX, Linux, Microsoft Office.